Mapping gene networks across breast cancer subtypes: testing their links to patient survival

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Summary

To compare gene networks between groups of patients, researchers usually build one network per group and compare the strongest connections. We asked how much of what that finds is real biology and how much is noise from unequal group sizes and mixed-up tissue samples. We expected a careful comparison to keep some real differences, but far fewer than a quick analysis suggests. We built GENIE3 networks for three breast cancer subtypes, gave every network the same number of patients, removed fat, stromal, and immune signals, and only called a connection different if it beat a null that re-ran the whole pipeline on shuffled labels. Basal tumors showed 45–51 real differences against the luminal subtypes, while the two luminal subtypes showed none. When we froze our findings and tested them in the independent METABRIC cohort, the gene modules held together, but their links to patient survival did not. Careful comparison keeps the real differences and cuts the false ones an easier method would report.

Introduction

Every cell in the body switches genes on and off, and the system that decides which genes affect which others is called a gene regulatory network. Breast cancer is not one disease. It comes in molecular subtypes (Luminal A, Luminal B, Basal, and HER2-enriched) that behave differently and respond to treatment differently. A natural question is whether these subtypes are regulated differently too—whether the same genes are controlled by different switches.

Computer tools like GENIE3 can infer these networks from gene expression data, so comparing subtypes should be as easy as building one network per subtype and seeing where they differ. In practice it is easy to do this badly, for three reasons.

First, the subtypes are different sizes. In the TCGA breast cancer data, Luminal A has 421 tumors and Basal has only 141. A network built on more patients simply looks sharper, so comparing them can confuse "more data" with "more biology." Second, a tumor sample is never pure tumor. It also contains fat, connective tissue, and immune cells, and the most variable genes can describe how much fat was in the sample rather than how the cancer cells are regulated. Third, calling an edge "different" between two subtypes needs a baseline for what random differences look like, and the easy baselines make chance differences look real.

So our question was simple. If we fix these three problems, which differences between the subtype networks are actually real? And do the stable parts of the networks tell us anything about

how patients do? We expected real differences between Basal and luminal tumors to survive, but far fewer than a naive comparison suggests. We also expected the network structure to carry over to a new patient cohort, even where survival signals did not.

Results

The subtypes' top edges overlap less across the luminal–basal divide

The two luminal subtypes shared about three times as many top edges with each other as either shared with Basal (top-edge overlap score: LumA–LumB 0.248, Basal–LumA 0.082, Basal–LumB 0.072; Figure 1).

Real differences cluster on Basal, and the luminal subtypes look alike

After the full permutation test, 45 edges differed between Basal and LumA (32 favoring LumA) and 51 between Basal and LumB (35 favoring LumB) at a 5% false discovery rate. No edge differed between LumA and LumB (Figure 2, Table 1).

The most reliable hubs are known basal markers

Nine to twelve genes per subtype stayed in the top-20 regulators in at least 70% of 30 bootstrap networks (Figure 3). They are mostly basal epithelial genes—*SOX10* (found in every bootstrap in all three subtypes), *KRT5*, *KRT14*, *KRT6B*, *KRT17*, *DSG3*, *DSC3*, *S100A7*, *KLK5*, and *KLK6*. No fat-cell genes survived, which shows the composition adjustment worked.

Stable modules are small shared cores, not clean subtype programs

Eight modules passed our stability cutoff (Figure 4): three in Basal (8, 6, and 6 genes), two in LumA (12 and 9), and three in LumB (12, 8, and 6). The strongest LumA and LumB modules were actually the same gene set, and Basal's top module shared two-thirds of its genes with it. The dominant coherent program is a shared epithelial core, not something unique to one subtype.

TCGA survival links were weak

Of the eight modules tested in TCGA (810 patients, 104 deaths), only LumA_M2 was significant on its own (hazard ratio 0.81, $p = 0.038$), and nothing survived correction for multiple testing

after adjusting for age, stage, and subtype. Three modules were frozen for validation: LumA_M2, Basal_M1, and LumB_M3 (Figure 5).

Modules held up in METABRIC, but the survival links did not

Both testable modules showed strong within-module correlation in METABRIC, far above random gene sets (Basal_M1 $z = 14.7$, LumA_M2 $z = 7.6$, both $p < 0.005$; LumB_M3 lost too many genes in mapping and was dropped). But neither survival association replicated in 1,892 patients with 728 events: LumA_M2 HR 1.00 ($p = 0.93$), Basal_M1 HR 0.99 ($p = 0.73$) (Figure 6, Table 2).

Discussion

Our main finding is that Basal breast tumors really are wired differently from luminal ones. That survived every control we applied. Luminal A and Luminal B, on the other hand, look the same at the level of individual edges, no matter what a quick edge comparison suggests. The real differences are few (45–51 edges) next to what a weaker null reports, which means most of what an easy analysis finds is noise. The stable hubs also make biological sense: they are basal epithelial markers, and the absence of fat genes shows the composition cleanup did its job.

The most instructive result is a negative one. The modules clearly exist as real co-expression units, since they transferred to a completely different platform and cohort, but they are not prognostic. METABRIC has seven times as many outcome events as TCGA, so the failure to replicate is not a small sample problem. The TCGA survival signal was never reliable. That is exactly what the frozen design is for. A study that peeked at METABRIC before fixing its modules could easily have produced a confirmed biomarker, and nobody would know it was false.

Our work has limits. Edge-by-edge testing misses dense programs: when twenty genes share a regulatory program, each edge carries only a small part of the signal, so a subtype-specific program can exist without any single edge looking significant. GENIE3 edges are statistical predictions, not proof that a regulator physically controls a target. Our composition correction used marker genes rather than a full deconvolution method, so some stromal signal likely remains. HER2-enriched tumors were excluded because there were too few for matched comparison. Finally, the survival analysis is observational: it links module activity to outcome but cannot say the module causes it.

Next steps would be to check whether the differential edges also appear in protein or chromatin data, to try the same procedure in another cancer type, and to test whether matched subsampling

changes conclusions in previously published network comparisons.

Materials and Methods

Data

TCGA-BRCA expression data (Illumina HiSeq RNA-seq) was downloaded from the UCSC Xena hub: 1,097 primary tumors, 841 with a PAM50 subtype call (LumA 421, LumB 192, Basal 141, HER2 67, Normal 23). METABRIC microarray data was fetched through the cBioPortal API for the genes selected in stage 1 (452 of 500 mapped). Subtypes with fewer than 100 samples were excluded, leaving LumA, LumB, and Basal. All pipeline code, configuration files, tests, and the synthetic validation cohorts are publicly available [8].

Stage 1: preprocessing

Genes with very low expression were removed. Adipocyte, stromal, and immune marker scores were regressed out of the matrix, and the 500 most variable genes were kept (444 of these overlapped the list selected without adjustment).

Stage 2: matched-size networks

GENIE3 (random forests, 100 trees) was run on subsamples matched to the smallest subtype ($n = 141$), averaged over 5 replicate fits per subtype so no subtype got a sharper network just for being bigger.

Stage 3: differential edges

For each subtype pair, each top edge's score difference was compared to a null distribution built by shuffling subtype labels and re-running the entire pipeline 10 times. Edges were called different at a 5% false discovery rate.

Stage 4: bootstrap stability

Each subtype network was refit on 30 bootstrap resamples (80% of patients each). A hub is a gene in the top-20 regulators in at least 70% of bootstraps. Modules are clusters of genes that keep appearing together across the bootstrap networks; only modules with coherence above 0.5 moved on to survival analysis.

Stage 5: survival analysis

Each patient's module activity was scored as the average within-sample rank of the module's genes, which lets the score transfer between RNA-seq and microarray platforms. We used Kaplan–Meier curves, log-rank tests, and Cox models (alone and adjusted for age, stage, and subtype) for overall survival. The module list was frozen at this point.

Stage 6: validation in METABRIC

The frozen modules were carried into METABRIC unchanged. Preservation was tested by comparing each module's mean within-module correlation against random gene sets of the same size. The Cox models were then re-run in 1,892 patients with 728 outcome events. Nothing from the discovery stage was edited.

Checking the pipeline

The pipeline is guarded by 125 automated tests that check, among other things, that planted regulators are found, that the permutation test is calibrated (uniform p-values when there is no real difference), that bootstrap hubs and modules recover planted signal, and that a synthetic two-cohort dataset with a known answer is analyzed correctly end to end.

References

1. Huynh-Thu VA, Irrthum A, Wehenkel L, Geurts P. Inferring regulatory networks from expression data using tree-based methods. *PLoS ONE*. 2010;5(9):e12776.

2. Parker JS, et al. Supervised risk predictor of breast cancer based on intrinsic subtypes. *J Clin Oncol*. 2009;27(8):1160–1167.

3. Cancer Genome Atlas Network. Comprehensive molecular portraits of human breast tumours. *Nature*. 2012;490:61–70.

4. Curtis C, et al. The genomic and transcriptomic architecture of 2,000 breast tumours reveals novel subgroups. *Nature*. 2012;486:346–352.

5. Yoshihara K, et al. Inferring tumour purity and stromal and immune cell admixture from expression data. *Nat Commun*. 2013;4:2612.

6. Marbach D, et al. Wisdom of crowds for robust gene network inference. *Nat Methods*. 2012;9:796–804.

7. Ideker T, Krogan NJ. Differential network biology. *Mol Syst Biol*. 2012;8:565.

8. Gupta S. genie3_brca_project: source code, tests, and data for this study. GitHub. 2026. `github.com/guptashivengupta-source/genie3_brca_project`

Acknowledgements

This independent study was carried out in the IDEA Lab at the University of North Texas under the guidance of Dr. Sanjukta Bhowmick. All code, tests, and synthetic validation cohorts are available at `github.com/guptashivengupta-source/genie3_brca_project`.

Figures and Tables

Table 1: Top-edge overlap and differential edges at a 5% false discovery rate.

Comparison	Top-edge Jaccard	Edges at FDR 5%
Basal vs LumA	0.082	45 (32 favor LumA)
Basal vs LumB	0.072	51 (35 favor LumB)
LumA vs LumB	0.248	0

Table 2: Hazard ratios for overall survival in discovery (TCGA) and validation (METABRIC).

Module	TCGA HR (p)	METABRIC HR (p)	Same dir.?	Replicated?
LumA_M2	0.81 (0.038)	1.00 (0.93)	no	no
Basal_M1	0.84 (0.074)	0.99 (0.73)	yes	no

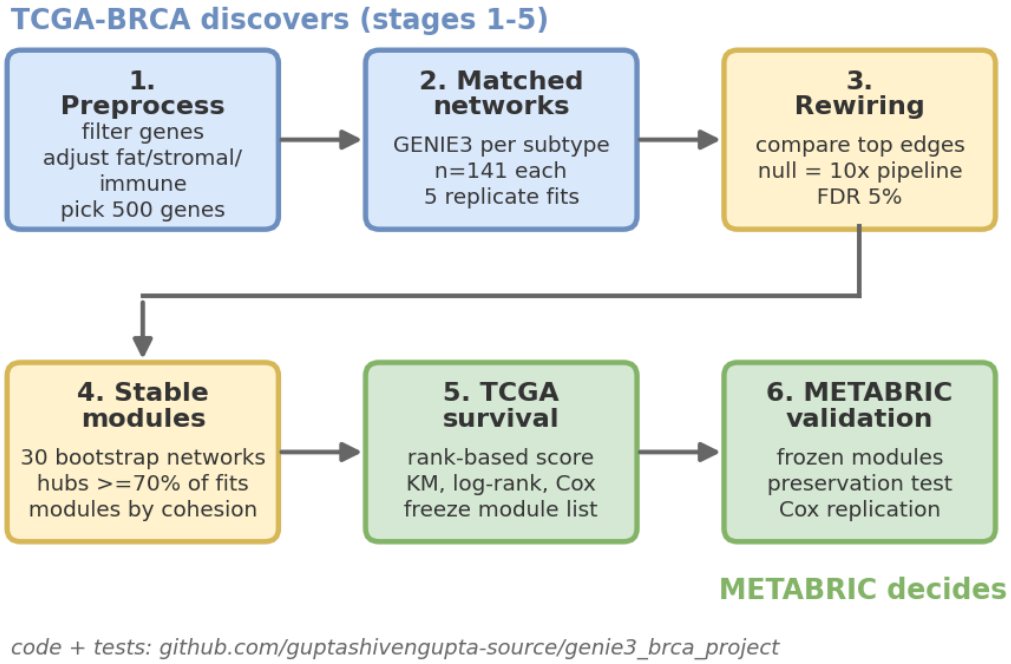


Figure 1: The six-stage pipeline. Stages 1–5 run entirely on TCGA-BRCA and are frozen before stage 6 tests the independent METABRIC cohort.

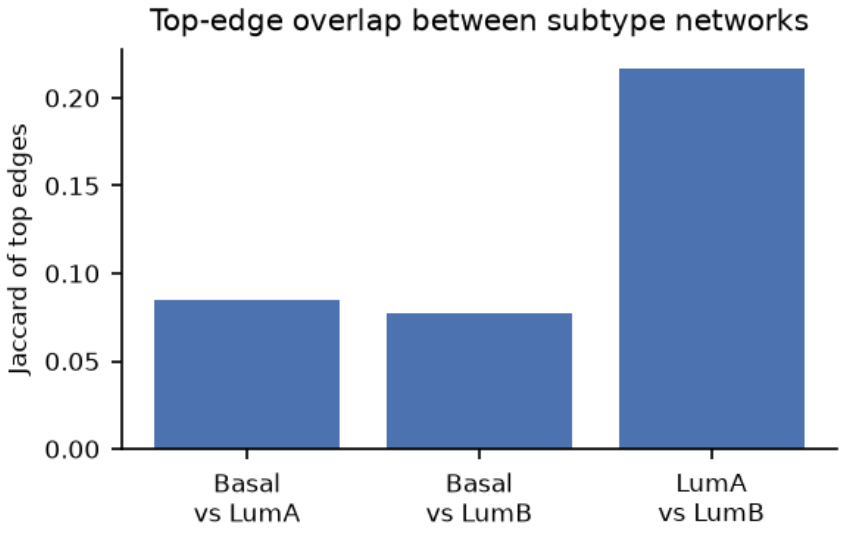


Figure 2: Top-edge overlap between subtype networks. Luminal subtypes share far more edges with each other than either shares with Basal.

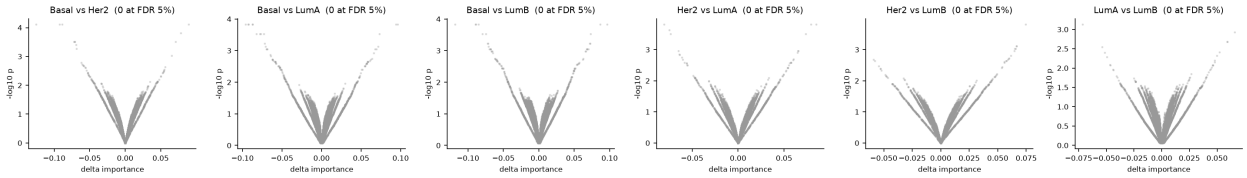


Figure 3: Differential edges for each subtype pair, colored by the subtype each edge favors. The LumA–LumB panel is empty at a 5% false discovery rate.

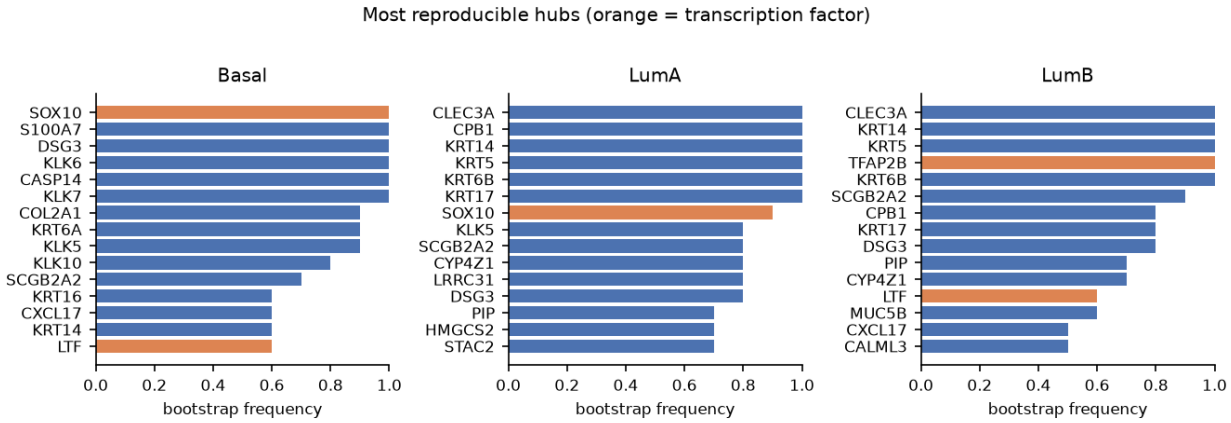


Figure 4: How often each gene appears as a top regulator across 30 bootstrap networks. Orange marks transcription factors.

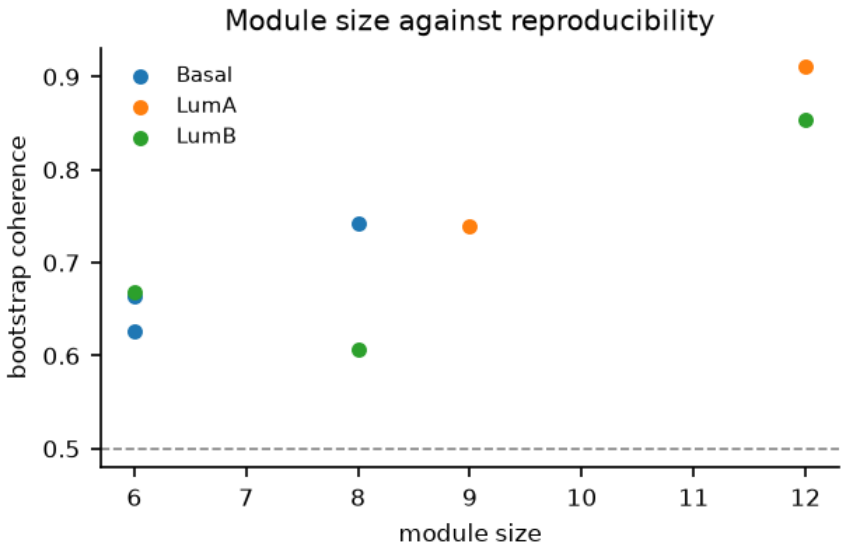


Figure 5: Module coherence (how often genes cluster together across bootstraps) and module size.

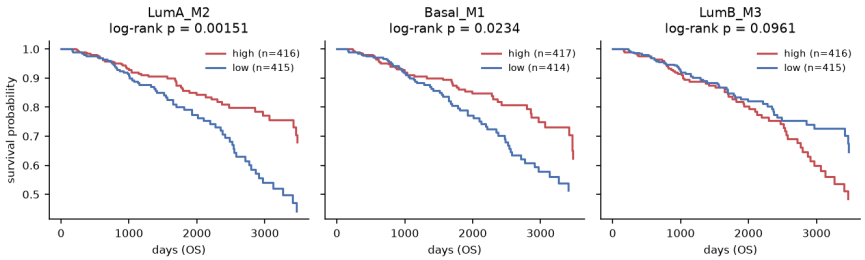


Figure 6: Kaplan–Meier survival curves for the frozen modules in TCGA.

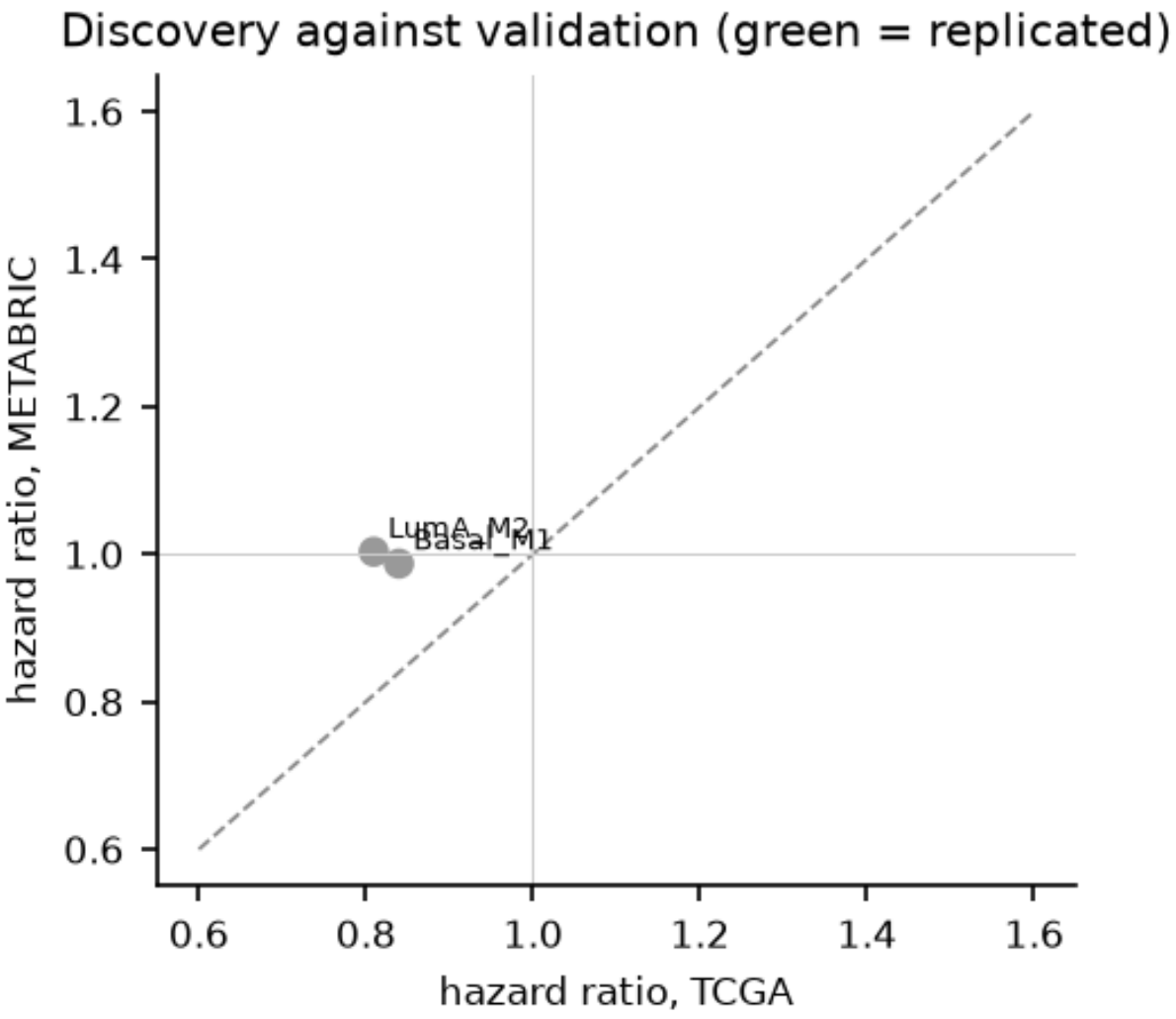


Figure 7: Module preservation and hazard ratios in discovery (TCGA) versus validation (METABRIC).